



Group 19: Clinic Appointment System
Phase 2

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Group Information

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1. Conceptual Design

1.1 Business rules

The following rules for database design are drawn from system operations and expressed in formal logic.

Appointment Booking Rules

- A patient can book multiple appointments, but each appointment must belong to exactly one patient.
- Only one staff member (nurse or doctor) may be allocated to each appointment.
- Every appointment needs to have a valid time and date.
- A patient has the option to reschedule or cancel an appointment.

Time Allocation Rules

- A time slot becomes available when an appointment is cancelled.
- Every appointment has a single time slot assigned to it.
- A daily limit of 50 appointments can be accommodated by the clinic.

Patient Management Rules

- A student patient must authenticate using their student ID before booking or accessing records
- It is not permitted to have duplicate patient records.
- Every patient has access to their personal records.

Staff Management Rules

- Every employee needs to have a role.
- Every doctor needs to have a license number and a specialization.
- Every employee needs to have specific working hours.

Walk-in Patient Rules

- Walk-in patients are allowed without prior booking.
- Walk-in patients must be given an available time slot.

Queue and Check-in Rules

- A check-in record is required for every appointment.
- Every consultation needs to have a start and end time.

System Integrity Rules

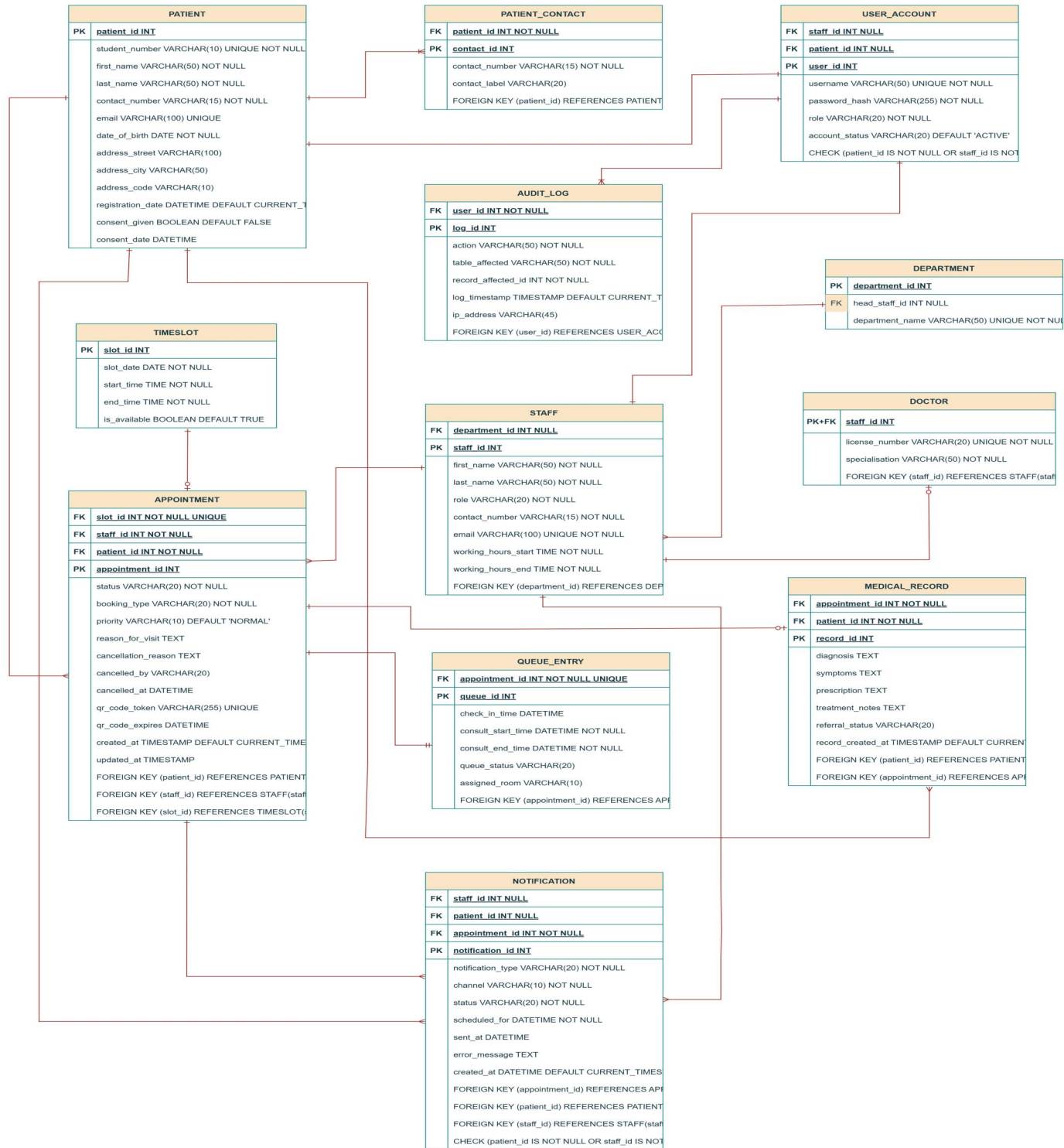
- All appointments must be validated before confirmation.

- Both the patient and the appointment must be connected to the medical records.

Notification Rule

- A reminder must be sent out 24 hours prior to each scheduled appointment.

1.2 Entity Relationship (ER) diagram



1.3 Relationships Between Our Tables

PATIENT → PATIENT_CONTACT

One-to-Many (1:M)

An individual patient may register several contact numbers, yet each contact entry corresponds to a single patient only. The foreign key attribute patient_id is stored within the PATIENT_CONTACT table.

Participation:

PATIENT_CONTACT depends entirely on PATIENT (a contact record cannot exist without a patient)

PATIENT may exist independently of PATIENT_CONTACT (a patient need not provide multiple contacts)

PATIENT → USER_ACCOUNT

One-to-One (1:1)

A patient may be linked to at most one portal account. The connection is established through the patient_id foreign key situated in the USER_ACCOUNT table.

Participation:

USER_ACCOUNT requires a valid PATIENT reference when associated with a patient

PATIENT participation is optional (account creation occurs separately from registration)

USER_ACCOUNT → AUDIT_LOG

One-to-Many (1:M)

A single user account may produce numerous audit entries as actions are performed within the system. Each log entry references exactly one user account via the user_id foreign key in the AUDIT_LOG table.

Participation:

AUDIT_LOG depends completely on USER_ACCOUNT (every log requires an actor)

USER_ACCOUNT may exist without logs (newly created accounts have no history)

DEPARTMENT → STAFF

One-to-Many (1:M)

A department may contain several staff members, whereas each staff member belongs to at most one department. The department_id foreign key resides in the STAFF table.

Participation:

STAFF association with DEPARTMENT is optional (personnel may be unassigned)

DEPARTMENT may exist without STAFF (a newly formed department may be empty)

STAFF → DEPARTMENT (heads)**One-to-Zero-or-One (1:0:1)**

A staff member may be designated as the head of a department, holding this position for zero or one department. The head_staff_id foreign key is placed within the DEPARTMENT table.

Participation:

STAFF participation is optional (the majority of staff do not head departments)

DEPARTMENT participation is optional (leadership positions may remain vacant)

STAFF → DOCTOR**One-to-Zero-or-One (1:0:1)**

A staff member may optionally hold the role of doctor. The DOCTOR table utilizes staff_id as both its primary identifier and foreign reference.

Participation:

DOCTOR requires an associated STAFF record (every doctor is a staff member)

STAFF may exist without a DOCTOR record (nurses and administrators are not doctors)

STAFF → USER_ACCOUNT**One-to-One (1:1)**

A staff member may possess at most one system account. The staff_id foreign key is maintained within the USER_ACCOUNT table.

Participation:

STAFF participation is optional (not all personnel require system access)

USER_ACCOUNT participation is optional (accounts may alternatively belong to patients)

STAFF → APPOINTMENT**One-to-Many (1:M)**

A staff member may be scheduled for numerous appointments, while each appointment is assigned to precisely one provider. The staff_id foreign key is stored in the APPOINTMENT table.

Participation:

APPOINTMENT requires a STAFF assignment (every booking needs a provider)

STAFF may have no APPOINTMENT records (newly hired personnel)

TIMESLOT → APPOINTMENT**One-to-One (1:1)**

Each time slot accommodates at most one appointment, and every appointment occupies exactly one time slot. The slot_id foreign key in APPOINTMENT carries a UNIQUE constraint.

Participation:

APPOINTMENT depends on TIMESLOT (a booking must have a scheduled time)

TIMESLOT may remain unoccupied (available slots await booking)

APPOINTMENT → QUEUE_ENTRY**One-to-One Mandatory (1:1)**

Every appointment generates exactly one queue entry upon patient arrival. The appointment_id foreign key in QUEUE_ENTRY includes a UNIQUE constraint.

Participation:

QUEUE_ENTRY cannot exist without APPOINTMENT (weak entity)

APPOINTMENT must have a corresponding QUEUE_ENTRY (check-in is required)

APPOINTMENT → MEDICAL_RECORD**One-to-Zero-or-One (1:0:1)**

An appointment may result in the creation of a medical record, limited to a single record per visit. The appointment_id foreign key is placed in MEDICAL_RECORD.

Participation:

MEDICAL_RECORD requires an APPOINTMENT (documentation tied to specific visit)

APPOINTMENT may lack a MEDICAL_RECORD (missed appointments generate no documentation)

PATIENT → APPOINTMENT**One-to-Many (1:M)**

A patient may schedule multiple appointments throughout their history with the clinic. The patient_id foreign key resides in the APPOINTMENT table.

Participation:

APPOINTMENT requires a PATIENT (every booking needs a patient)

PATIENT may have no APPOINTMENT records (newly registered individuals)

PATIENT → MEDICAL_RECORD**One-to-Many (1:M)**

A patient may accumulate numerous medical records spanning multiple visits. The patient_id foreign key is stored in the MEDICAL_RECORD table.

Participation:

MEDICAL_RECORD requires a PATIENT (documentation belongs to an individual)

PATIENT may have no MEDICAL_RECORD entries (no prior visits)

APPOINTMENT → NOTIFICATION**One-to-Many (1:M)**

An appointment may trigger several notifications throughout its lifecycle. The appointment_id foreign key is placed in the NOTIFICATION table.

Participation:

NOTIFICATION requires an APPOINTMENT (messages tied to specific booking)

APPOINTMENT may have no NOTIFICATION records (messages generated as required)

PATIENT → NOTIFICATION**One-to-Many (1:M)**

A patient may receive numerous notifications from the system. The patient_id foreign key resides in the NOTIFICATION table.

Participation:

NOTIFICATION may reference a PATIENT or alternatively a STAFF member

PATIENT may have no NOTIFICATION entries

STAFF → NOTIFICATION**One-to-Many (1:M)**

A staff member may receive multiple notifications regarding schedule updates and appointments. The staff_id foreign key is stored in the NOTIFICATION table.

Participation:

NOTIFICATION may reference a STAFF member or alternatively a PATIENT

STAFF may have no NOTIFICATION entries

1.4 Derived Attributes

PATIENT Table

The stored date_of_birth value enables dynamic calculation of the patient's age whenever required. Computing age at query time eliminates the maintenance burden of updating a static age column annually.

QUEUE_ENTRY Table

The duration a patient spends waiting is determined by calculating the interval between consult_start_time and check_in_time. This derived measurement supports the clinic's objective of maintaining wait times under fifteen minutes without requiring additional storage.

APPOINTMENT Table

Certain appointment states may be inferred from contextual data rather than stored explicitly. When the scheduled time passes without an associated check-in record, the appointment can be interpreted as a no-show without necessitating a dedicated status update.

1.5 Normalization Description

First Normal Form (1NF)

Every attribute throughout the schema holds atomic values exclusively. The PATIENT table decomposes address information into separate columns for street, city, and postal code rather than storing concatenated text. The PATIENT_CONTACT entity addresses the multi-valued nature of telephone numbers by allocating a dedicated row for each contact entry. No repeating groups appear within any table structure.

Second Normal Form (2NF)

All tables within the design utilize single-column surrogate primary keys. This approach eliminates any possibility of partial dependency since composite keys do not exist within the schema. Where candidate keys appear, such as student_number in PATIENT, all non-key attributes remain fully dependent on the primary key alone.

Third Normal Form (3NF)

The schema contains no transitive dependencies. The DOCTOR table demonstrates this principle by relocating license_number and specialisation away from the STAFF table. These attributes depend

upon the condition of holding a doctor role rather than upon the staff identifier itself. Isolating these fields in a subtype table prevents update anomalies and removes the necessity for NULL entries among non-doctor personnel. Similarly, the APPOINTMENT table references provider details through foreign keys rather than duplicating attributes such as physician names, ensuring that modifications to staff information occur in a single location.

```

    erDiagram
        PATIENT ||--}| PATIENT_CONTACT : "1:M"
        PATIENT_CONTACT ||--}| USER_ACCOUNT : "1:M"
        USER_ACCOUNT ||--}| AUDIT_LOG : "1:M"
        AUDIT_LOG ||--}| STAFF : "1:M"
        STAFF ||--}| DEPARTMENT : "1:M"
        STAFF ||--}| DOCTOR : "1:M"
        STAFF ||--}| APPOINTMENT : "1:M"
        STAFF ||--}| QUEUE_ENTRY : "1:M"
        STAFF ||--}| NOTIFICATION : "1:M"
        APPOINTMENT ||--}| TIMESLOT : "1:M"
        APPOINTMENT ||--}| MEDICAL_RECORD : "1:M"
        MEDICAL_RECORD ||--}| QUEUE_ENTRY : "1:M"
        MEDICAL_RECORD ||--}| NOTIFICATION : "1:M"

        PATIENT {
            string patient_id PK
            string student_number UNIQUE
            string first_name VARCHAR(50)
            string last_name VARCHAR(50)
            string contact_number VARCHAR(15)
            string email VARCHAR(100) UNIQUE
            datetime date_of_birth
            string address_street VARCHAR(100)
            string address_city VARCHAR(50)
            string address_code VARCHAR(10)
            datetime registration_date
            boolean consent_given DEFAULT FALSE
            datetime consent_date
        }
        PATIENT_CONTACT {
            string contact_id PK
            string patient_id FK
            string contact_number VARCHAR(15)
            string contact_label VARCHAR(20)
        }
        USER_ACCOUNT {
            string user_id PK
            string patient_id FK
            string username VARCHAR(50) UNIQUE
            string password_hash VARCHAR(255)
            string role VARCHAR(20)
            string account_status VARCHAR(20)
            string CHECK {patient_id IS NOT NULL OR staff_id IS NOT NULL}
        }
        AUDIT_LOG {
            string log_id PK
            string user_id FK
            string action VARCHAR(50)
            string table_affected VARCHAR(50)
            int record_affected_id
            timestamp log_timestamp
            string ip_address VARCHAR(45)
        }
        STAFF {
            string staff_id PK
            string department_id FK
            string first_name VARCHAR(50)
            string last_name VARCHAR(50)
            string role VARCHAR(20)
            string contact_number VARCHAR(15)
            string email VARCHAR(100) UNIQUE
            datetime working_hours_start
            datetime working_hours_end
        }
        DEPARTMENT {
            string department_id PK
            string head_staff_id FK
            string department_name VARCHAR(50) UNIQUE
        }
        DOCTOR {
            string staff_id PK FK
            string license_number VARCHAR(20) UNIQUE
            string specialisation VARCHAR(50)
        }
        TIMESLOT {
            string slot_id PK
            datetime slot_date
            datetime start_time
            datetime end_time
            boolean is_available DEFAULT TRUE
        }
        APPOINTMENT {
            string appointment_id PK
            string slot_id FK
            string staff_id FK
            string patient_id FK
            string status VARCHAR(20)
            string booking_type VARCHAR(20)
            string priority VARCHAR(10) DEFAULT 'NORMAL'
            string reason_for_visit TEXT
            string cancellation_reason TEXT
            datetime cancelled_by VARCHAR(20)
            datetime cancelled_at
            string qr_code_token VARCHAR(255) UNIQUE
            datetime qr_code_expires
            datetime created_at
            datetime updated_at
        }
        QUEUE_ENTRY {
            string queue_id PK
            string appointment_id FK
            datetime check_in_time
            datetime consult_start_time
            datetime consult_end_time
            string queue_status VARCHAR(20)
            string assigned_room VARCHAR(10)
        }
        NOTIFICATION {
            string notification_id PK
            string staff_id FK
            string patient_id FK
            string appointment_id FK
            string notification_type VARCHAR(20)
            string channel VARCHAR(10)
            string status VARCHAR(20)
            datetime scheduled_for
            datetime sent_at
            string error_message TEXT
            datetime created_at
        }
        MEDICAL_RECORD {
            string record_id PK
            string appointment_id FK
            string patient_id FK
            string diagnosis TEXT
            string symptoms TEXT
            string prescription TEXT
            string treatment_notes TEXT
            string referral_status VARCHAR(20)
            datetime record_created_at
        }
    
```